

Highlights

When Conformal Fusion Cannot Be Calibrated: Attainability Limits for Trace-Based Detection of Prompt-Injection Compromise

- Conformal p-value floors cap the evidence any e-value calibrator can express
- Dependence-robust conformal fusion is inert below 312 calibration traces
- Eight merging rules rank alike but differ hugely in a-priori false-alarm control
- Bonferroni alone fires; injection evidence is sparse in the tail, dense in bulk
- Subtracting the benign maximum provably caps AUROC at one half